Fuzz Campaign Physical-Core Parallelism

Tactix Development Project

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Outcome

The active consolidation fuzz campaign is correctness-clean so far but was submitted at the wrong parallel width. Claude Code's ownership audit reports six generated/stress legs as independent `-c 2` jobs with two workers each. On the cluster's SMT-2 nodes this is approximately one physical core per leg, or about six physical cores at peak. The campaign has no arrays or hidden throttle and never achieved the intended 50-physical-core parallelism.

This affects wall time, not seeds, case count, coverage, or oracle semantics. At discovery the two remaining legs were already at 8,700/10,000 and 9,000/10,000 cases with zero failures. They are retained to completion rather than cancelled and repeated. No new fuzz campaign was submitted.

The generic fuzz wrapper is corrected so its default "50" means 50 physical cores and 50 workers, rather than 50 logical CPUs and 50 workers sharing 25 physical cores. The Development Plan now records the same contract for future Claude-owned generated fuzz legs.

1 Observed campaign shape

Claude reports prep jobs `1648187`/`1648211`, generated/stress legs `1648214`–`1648218`, and reach job `1648219`. Each generated leg requests two logical CPUs and 16 GiB, and launches two workers. Four legs are terminal clean; Congruence-stress and Sweep-stress remain the two original in-flight legs. Claude, not Codex, inspected these job records and fuzz artifacts, consistent with Development Plan §8.2.

The prior `tools/fuzz/slurm.sh` default requested `-cpus-per-task=50` with `-jobs 50`. Because the nodes expose two hardware threads per physical core, that generic shape reserved only 25 physical cores. The active custom campaign did not use even that wrapper and instead used its two-worker leg shape.

2 Validated future contract

Allocation-only job `1648629` ran no solver and no fuzz case. It completed 0:0 and reports:

Resource	Observed value
Logical CPUs	100
Scheduler memory	400,000 MiB
Allowed CPU mask	`0-49,96-145`
Physical cores represented	50

The mask contains cores 0–49 and their SMT siblings 96–145, proving that 100 logical CPUs reserve 50 physical cores without relying on a scheduler hint. Memory is 50 times the partition's effective 8,000 MiB per physical core. Future generated fuzz legs launch 50 workers, leaving one worker per physical core while the allocation reserves both siblings.

The wrapper now accepts `-physical-cores P`, derives `cpus-per-task=2P`, `mem=8000P MiB`, and `jobs=P`, and rejects values outside 1–96. A mocked submission test verifies the default 100/400,000/50 shape; a three-core control verifies 6/24,000/3 scaling; zero is rejected before submission. Shell syntax validation passes.

Preparation, small fixed-corpus work, or intrinsically narrower reachability probes may use less only with a recorded reason. A generated fuzz leg may not silently fall back to one or two workers. This is parallelism inside one fuzz campaign, not the three-wave repetition used for decision-grade performance timing.

3 Next action

Let Claude finish and audit the two nearly complete legs of the existing single campaign. A terminal PASS authorizes the already-complete consolidation gate to proceed to its own final report, commit, and push; a terminal failure enters the mandatory bounded Claude repair workflow. The corrected 50-physical-core wrapper applies to the next required fuzz campaign and does not retroactively invalidate the current campaign's completed coverage.